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Theory and Methodology

Lower-bounding and heuristic methods for a refuse collection vehicle routing problem ☆

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Abstract

A set of routes that minimizes the total collecting cost of the household refuse in a quarter of Lisbon may be obtained solving a Capacitated Arc Routing Problem (CARP) with side constraints. The CARP is known to be an NP-hard problem. We present two lower-bounding methods, both based on the transportation model, in which we have been able to incorporate some of the side constraints. We also present a three-phase heuristic to generate a near-optimal solution from the solution obtained with the first lower-bounding method. For the relative gap between the heuristic solution value and its associated lower bound value we give a theoretical worst-case bound and computational experience obtained with a set of test problems. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

Collecting household refuse is one of the most difficult operational problems faced by local authorities in any large city. For the moment, we are concerned with the optimization of the refuse collection vehicle tours for a particular quarter of Lisbon, since in Lisbon the refuse collection is planned quarter by quarter.

The operating system, at a specific quarter, may be roughly described as follows: each vehicle departs from a central garage, located outside the quarter, and goes directly to a crew base, inside the quarter, where the crew (apart from the driver) is waiting for its arrival to initiate their working period. The first trip begins at the crew base and the vehicle collects refuse along the streets, up to the point where the garbage to be collected along the next street segment would exceed the vehicle's residual capacity. From this point the vehicle goes directly to a recycling facility, located well away from the quarter, to unload the refuse. The

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vehicle then goes back to the quarter and begins its second trip. Every time (except for the last) the vehicle's residual capacity becomes insufficient for the next street segment it must head to the recycling facility, to unload, and then return to the quarter. After completing its final trip, the vehicle first goes to the crew base, leaving the crew at its pick up point, then proceeds to the recycling facility to unload for the last time, and finally returns to the central garage. Fig. 1 illustrates this operating system, for one vehicle.

The shadowed area (Fig. 1) represents the hard part of the problem to solve. At the beginning of the journey the vehicle will simply follow the shortest path from the garage to the crew base. At the end of the journey, from the crew base to the recycling facility and from the recycling facility back to the garage, the vehicle will once again follow the respective shortest paths. These three shortest paths are represented by the thick arcs in the figure.

By a vehicle's *tour* we mean the total course driven since departing from the crew base point, until returning back there to drop the crew. Each tour is a sequence of two or more *trips*: the *initial trip*, from the crew base to the recycling facility, the *final trip* from the recycling facility back to the crew base and the *intermediate trips* (if any) from the recycling facility back to it.

Since the number of vehicles in the fleet is scarce for the total demand, all vehicles must be used and each vehicle performs at least two trips. The recycling facility is far away from the quarter and there is no refuse to be collected in the streets adjacent to it.

A set of tours that minimizes the total cost of collecting the household refuse in the quarter may be obtained by solving a Capacitated Arc Routing Problem (CARP) with some additional constraints. The CARP is known to be an NP-hard problem [8].

As shown in two recent surveys, Assad and Golden [1], Eiselt et al. [5,6], arc routing problems may be used to solve a huge number of real world problems in very different domains (winter gritting, postal distribution, meter reading, street sweeping, etc.).

We present two lower-bounding methods for the refuse collection problem, both based on the transportation model. In these transportation problems we have incorporated some additional constraints not considered in the lower bound presented by Beltrami and Bodin [2].

We have also developed a three-phase heuristic to generate a near-optimal solution from the solution obtained with the first lower-bounding method, similarly and in the line of the ideas proposed by Bodin [3]; by Bodin and Kursh [4] and by Frederickson et al. [7] for some routing problems. The value of this feasible solution represents an upper bound to our problem.

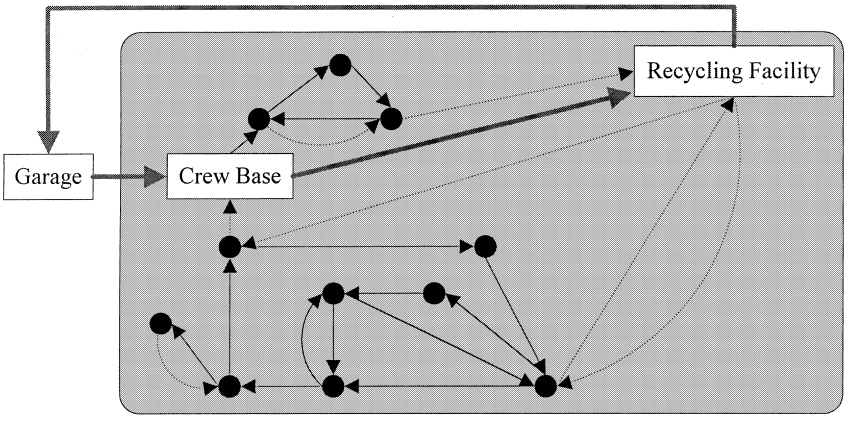


Fig. 1. The problem.

We derived a theoretical worst-case bound for the relative gap between the heuristic solution value and its associated lower bound. Computational experience is presented for these two methods over a set of test problems.

2. Notation and problem formulation

2.1. Notation

Let $G = (V, A)$ be a directed graph, with $n + 1$ nodes ($|V| = n + 1$), representing the quarter and the recycling facility. Node 1 represents the crew base and node $n + 1$ the recycling facility. Remaining nodes, in the set $U = V \setminus \{1, n + 1\}$, represent street crossings or dead-end streets. Each arc, $(i, j) \in A$, represents a street segment in between two street crossings. A one way street is represented by an arc with the same orientation. A two-way street is represented by a pair of reverse arcs: (i, j) and (j, i).

We denote by

P	the number of vehicles used to collect the refuse in the quarter;
W	the vehicle capacity;
G_R	$= (V, A_R)$, the *demand graph*, with $A_R \subseteq A$ the set of arcs representing street segments requiring refuse collection, also called *demand arcs*. Arcs in $A \setminus A_R$, if any, may be used for *deadheading*;
$q_{i,j}$	the amount of refuse to be collected at $(i, j) \in A_R$; $Q_T = \sum_{(i,j)\in A_R} q_{i,j}$;
$c_{i,j}$	the cost (or time) of collecting refuse at $(i, j) \in A_R$;
C_{A_R}	$= \sum_{(i,j)\in A_R} c_{i,j}$ the *service cost*;
$d_{i,j}$	the cost (or time) of traversing $(i, j) \in A$ without collecting refuse, also called the *deadheading cost* ($d_{i,j} < c_{i,j},\ \forall (i, j) \in A_R$);
$spl(i, j)$	the deadheading cost of the minimum cost path from i ($i \in V$) to j ($j \in V$);
$scl(\ell)$	the deadheading cost of the minimum cost circuit that contains ℓ ($\ell \in V$). This circuit is formed by, at least, two arcs: (ℓ, v) and (v, ℓ), with $v \neq \ell$;
$lcl(\ell)$	$= \max_{i\in V\setminus\{\ell\}} \{spl(i, \ell) + spl(\ell, i)\},\ \ell \in V$;
$d^-(\ell)$	the indegree of node $\ell \in V$ in G_R, i.e., $d^-(\ell) = \lvert\{(i, \ell):\ (i, \ell) \in A_R\}\rvert$;
$d^+(\ell)$	the outdegree of $\ell \in V$ in G_R, i.e., $d^+(\ell) = \lvert\{(\ell, j):\ (\ell, j) \in A_R\}\rvert$;
$D(\ell)$	$= d^-(\ell) - d^+(\ell) \quad \forall\ \ell \in V$.

Consider all demand arcs ordered in decreasing order of their demands (i.e. $q_1 \geqslant q_2 \geqslant \cdots \geqslant q_m$) and let ℓ be such that

$$\sum_{a=1}^{\ell} q_a \leqslant W < \sum_{a=1}^{\ell+1} q_a,$$

then, in any optimal solution at most L trips will be performed, with $L = \lceil m/\ell \rceil$ and $m = |A_R|$.

2.2. Problem formulation

Assume that

- $r = 1, \ldots, P$ represent the P initial trips;
- $r = P + 1, \ldots, 2P$ represent the P final trips;
- $r = 2P + 1, \ldots, L$ represent the intermediate trips.

For $r = 1, \ldots, L$ let us define:

$$x_{i,j}^r = \begin{cases} 1 & \text{if the refuse in arc } (i,j) \text{ is collected during trip } r, \\ 0 & \text{otherwise,} \end{cases} \qquad \forall (i,j) \in A_R;$$

$y_{i,j}^r$ = the number of times that arc $(i,j) \in A$, is traversed during trip r, as a deadheading arc (i.e., without collecting refuse).

We may formulate the refuse collection problem as

$$\min\ z = \sum_{r=1}^{L} \sum_{(i,j)\in A_R} c_{i,j} x_{i,j}^r + \sum_{r=1}^{L} \sum_{(i,j)\in A} d_{i,j} y_{i,j}^r \tag{1}$$

s.t.:

$$\sum_{(i,\ell)\in A_R} x_{i,\ell}^r + \sum_{(i,\ell)\in A} y_{i,\ell}^r - \sum_{(\ell,i)\in A_R} x_{\ell,i}^r - \sum_{(\ell,i)\in A} y_{\ell,i}^r = 0 \quad \forall i \in U, \quad r = 1,\ldots,L, \tag{2}$$

$$\sum_{(1,\ell)\in A_R} x_{1,\ell}^r + \sum_{(1,\ell)\in A} y_{1,\ell}^r - \sum_{(\ell,1)\in A_R} x_{\ell,1}^r - \sum_{(\ell,1)\in A} y_{\ell,1}^r = 1, \quad r = 1,\ldots,P, \tag{3}$$

$$\sum_{(\ell,n+1)\in A} y_{\ell,n+1}^r = 1, \quad r = 1,\ldots,P, \tag{4}$$

$$\sum_{(n+1,\ell)\in A} y_{n+1,\ell}^r = 0, \quad r = 1,\ldots,P, \tag{5}$$

$$-\sum_{(1,\ell)\in A_R} x_{1,\ell}^r - \sum_{(1,\ell)\in A} y_{1,\ell}^r + \sum_{(\ell,1)\in A_R} x_{\ell,1}^r + \sum_{(\ell,1)\in A} y_{\ell,1}^r = 1, \quad r = P+1,\ldots,2P, \tag{6}$$

$$\sum_{(n+1,\ell)\in A} y_{n+1,\ell}^r = 1, \quad r = P+1,\ldots,2P, \tag{7}$$

$$\sum_{(\ell,n+1)\in A} y_{\ell,n+1}^r = 0, \quad r = P+1,\ldots,2P, \tag{8}$$

$$\sum_{(1,\ell)\in A_R} x_{1,\ell}^r + \sum_{(1,\ell)\in A} y_{1,\ell}^r - \sum_{(\ell,1)\in A_R} x_{\ell,1}^r - \sum_{(\ell,1)\in A} y_{\ell,1}^r = 0, \quad r = 2P+1,\ldots,L, \tag{9}$$

$$\sum_{(n+1,\ell)\in A} y_{n+1,\ell}^r - \sum_{(\ell,n+1)\in A} y_{\ell,n+1}^r = 0, \quad r = 2P+1,\ldots,L, \tag{10}$$

$$\sum_{(n+1,\ell)\in A} y_{n+1,\ell}^r \leqslant 1, \quad r = 2P+1,\ldots,L, \tag{11}$$

$$\sum_{\substack{(i,j)\in A_R \\ i,j\in S}} x_{i,j}^r + \sum_{\substack{(i,j)\in A \\ i,j\in S}} y_{i,j}^r \leqslant M \left(\sum_{\substack{(i,j)\in A_R \\ i\in S,\ j\notin S}} x_{i,j}^r + \sum_{\substack{(i,j)\in A \\ i\in S,\ j\notin S}} y_{i,j}^r + \sum_{\substack{(i,j)\in A_R \\ i\notin S,\ j\in S}} x_{i,j}^r + \sum_{\substack{(i,j)\in A \\ i\notin S,\ j\in S}} y_{i,j}^r \right) \quad \forall\ S \subseteq V \setminus \{n+1\}, \quad r = 1,\ldots,L, \tag{12}$$

$$\sum_{(i,j)\in A_R} q_{i,j}\, x^r_{i,j} \leqslant W, \quad r = 1,\ldots,L, \tag{13}$$

$$\sum_{r=1}^{L} x^r_{i,j} = 1 \quad \forall (i,j) \in A_R, \tag{14}$$

$$x^r_{i,j} \in \{0,1\} \quad \forall (i,j) \in A_R, \quad r = 1,\ldots,L, \tag{15}$$

$$y^r_{i,j} \geqslant 0 \text{ integer} \quad \forall (i,j) \in A, \quad r = 1,\ldots,L. \tag{16}$$

Conditions:

Eq. (2)	guarantee trip continuity for each trip in all nodes in U;
Eqs. (3)–(5)	define the initial trips as P trips from the crew base to the recycling facility;
Eqs. (6)–(8)	define the final trips as P trips from the recycling facility to the crew base;
Eqs. (9)–(11)	characterize intermediate trips as trips from the recycling facility and back to it;
Eq. (12)	with $M > 0$ being a very large constant in order to preclude the formation of illegal trips, i.e. trips that do not include node $n+1$;
Eq. (13)	impose that the refuse collected along each trip does not exceed the vehicle's capacity, W;
Eq. (14)	guarantee that each street is serviced exactly once in one and only one trip.

The solution of the above problem is a set of tours that minimizes the total cost of collecting the household refuse in the quarter considered.

This problem contains all the constraints that define a CARP [6] and some additional constraints to define the configuration of the trips and their number: initial trips correspond to P paths from node 1 to node $n+1$, final trips correspond to P paths from node $n+1$ to node 1 and intermediate trips correspond to at most $L-2P$ circuits from node $n+1$ and back to it.

3. Lower-bounding methods

Next we present two methods, based on formulation (1)–(16), to compute lower bounds for the optimal value of the refuse collection problem. A detailed proof of validity of both these methods can be found in Mourão [9].

3.1. Method LB1

In method LB1 we first generate a relaxation for problem (1)–(16) in which the set of all trips is aggregated. Based on this relaxation we are able to construct two easily solvable sub-problems and to obtain a valid lower bound.

Let $f_{i,j} = \sum_{r=1}^{L} y^r_{i,j}$ be the number of times that arc $(i,j) \in A$ is traversed as a deadheading arc in all trips. Note that

$$\sum_{r=1}^{L} \sum_{(i,\ell)\in A_R} x^r_{i,\ell} = d^+(i) \quad \text{and} \quad \sum_{r=1}^{L} \sum_{(\ell,i)\in A_R} x^r_{\ell,i} = d^-(i).$$

So, if we aggregate conditions for the set of trips, we get:

- from Eq. (2)

$$\sum_{(i,\ell)\in A} f_{i,\ell} - \sum_{(\ell,i)\in A} f_{\ell,i} = D(i) = d^-(i) - d^+(i) \quad \forall i \in U.$$

- from Eqs. (3), (6) and (9)

$$\sum_{(1,\ell)\in A} f_{1,\ell} - \sum_{(\ell,1)\in A} f_{\ell,1} = D(1).$$

- from Eqs. (4), (5), (7), (8) and (10)

$$\sum_{(n+1,\ell)\in A} f_{n+1,\ell} - \sum_{(\ell,n+1)\in A} f_{\ell,n+1} = 0 = D(n+1).$$

Note also that, from Eq. (14), $\sum_{r=1}^{L} \sum_{(i,j)\in A_R} q_{i,j} x_{i,j}^r = Q_T$. Thus, from Eq. (13) and knowing that L is integer, we get

$$L \geqslant \lceil Q_T/W \rceil.$$

The integer $\lceil Q_T/W \rceil$ represents a lower bound on the number of trips in any feasible solution. Since we have P initial trips and P final trips, $\lceil Q_T/W \rceil - 2P$ is a lower bound on the number of intermediate trips.

For each intermediate trip actually performed we get: $\sum_{(n+1,\ell)\in A} y_{n+1,\ell}^r = 1$, from conditions (11) and (12). So, for the set of all trips we have

$$\sum_{r=1}^{L} \sum_{(n+1,\ell)\in A} y_{n+1,\ell}^r \geqslant \lceil Q_T/W \rceil - P \equiv K \Longleftrightarrow \sum_{(n+1,\ell)\in A} f_{n+1,\ell} \geqslant K.$$

After performing these transformations we have two independent sub-problems: one on variables $x_{i,j}^r$ and another on variables $f_{i,j}$.

Sub-problem (A1)

$$\text{(A1)} \quad \min z_{A1} = \sum_{r=1}^{L} \sum_{(i,j)\in A_R} c_{i,j}\, x_{i,j}^r$$

s.t. :

$$\sum_{r=1}^{L} x_{i,j}^r = 1 \quad \forall (i,j) \in A_R; \tag{14}$$

$$x_{i,j}^r \in \{0,1\} \quad \forall (i,j) \in A_R;\ r = 1,\ldots,L. \tag{15}$$

This problem's optimal value is just the service cost C_{A_R}.

Sub-problem (B1)

$$\text{(B1)} \quad \min\ z_{B1} = \sum_{(i,j)\in A} d_{i,j} f_{i,j}$$

s.t. :

$$\sum_{(i,\ell)\in A} f_{i,\ell} - \sum_{(\ell,i)\in A} f_{\ell,i} = D(i) \quad \forall i \in V, \tag{17}$$

$$\sum_{(1,\ell)\in A} f_{1,\ell} \geqslant P - d^{+}(1), \tag{18}$$

$$\sum_{(n+1,\ell)\in A} f_{n+1,\ell} \geqslant K, \tag{19}$$

$$f_{i,j} \geqslant 0 \text{ integer} \quad \forall (i,j) \in A. \tag{20}$$

Using network flow terminology we may say that conditions (17) are the typical node balance constraints and that conditions (18) and (19) impose lower bounds on the flow out of nodes 1 and $n+1$.

In terms of the real world problem, conditions (17) account for route continuity at each node and constraints (18) and (19) represent the need of extra deadheading to form a tour for each vehicle (remember that the tours must begin at the crew base and must visit the recycling facility at least K times). Therefore, (18) and (19) represent tour configuration and capacity constraints, respectively, that are not present in the problem addressed in Ref. [2].

Without constraints (18) and (19) we would have a minimum cost flow problem on graph $G = (V, A)$. Following Beltrami and Bodin [2], this flow problem could be transformed into a transportation problem just by creating an origin for each node $i \in V$ with $D(i) > 0$ and a destination for each node $i \in V: D(i) < 0$. With constraints (18) and (19) we would need to have nodes as origins and destinations simultaneously and this is not allowed by the transportation model.

Since a minimum cost flow problem is easily solvable, we first transform (B1) into an equivalent flow problem (B1′). We can then show that (B1′) can be solved as a transportation problem (TP1).

Hereafter we assume that $D(1) > 0$ and $d^{-}(1) < P$ (the same rationale applies for the other cases). So node 1 and node $n+1$ are both, simultaneously, an origin and a destination of (B1).

Generically, the construction of problem (B1′) requires replicating all nodes representing simultaneously origins and destinations in (B1). One of the replica represents the initial node as the origin and the other as the destination.

In this line of reason, node $(n+1)'$ is a surrogate for node $n+1$ as an origin, with supply K, and $n+1$ represents node $n+1$ as a destination, with demand K. Node $1'$ is a surrogate for node 1 as an origin, with supply $P - d^{+}(1)$, and 1 represents node 1 as a destination, with demand $P - d^{-}(1)$.

Consider the graph $G' = (V', A')$ defined by

$$V' = V \cup \left\{1', (n+1)'\right\},$$

$$A' = \left[A \setminus \left(A^{+}_{n+1} \cup A^{+}_{1}\right)\right] \cup \left\{\left(n+1,\ (n+1)'\right)\right\} \cup \left\{\left(1,\ 1'\right)\right\} \cup A^{+}_{(n+1)'} \cup A^{+}_{1'},$$

where

$$A^{+}_{k} = \{(k,\ell) \in A:\ \ell \in V\}, \quad k \in V$$

and

$$A^{+}_{k'} = \left\{\left(k',\ell\right):\ (k,\ \ell) \in A \ \wedge\ \ell \in V\right\}, \quad k \in \{1,\ n+1\}.$$

Let $d'_{i,j}$ be the unit costs of arcs in A', given by

$$d'_{i,j} = \begin{cases} d_{u,v} & \text{if } (i,j) \in A' \text{ is a copy of } (u,v) \in A, \\ 0 & \text{otherwise.} \end{cases}$$

It can be shown that solving sub-problem (B1) is equivalent to solving

$$(\mathrm{B1}') \quad \min z_{\mathrm{B1}'} = \sum_{(i,j)\in A'} d'_{i,j} f'_{i,j} \tag{21}$$

s.t.:

$$\sum_{(i,\ell)\in A'} f'_{i,\ell} - \sum_{(\ell,i)\in A'} f'_{\ell,i} = D(i) \quad \forall\, i \in U, \tag{22}$$

$$\sum_{(\ell,n+1)\in A'} f'_{\ell,n+1} - \sum_{(n+1,\ell)\in A'} f'_{n+1,\ell} = K, \tag{23}$$

$$\sum_{((n+1)',\ell)\in A'} f'_{(n+1)',\ell} - \sum_{(\ell,(n+1)')\in A'} f'_{\ell,(n+1)'} = K, \tag{24}$$

$$\sum_{(1',\ell)\in A'} f'_{1',\ell} - \sum_{(\ell,1')\in A'} f'_{\ell,1'} = P - d^+(1), \tag{25}$$

$$\sum_{(\ell,1)\in A'} f'_{\ell,1} - \sum_{(1,\ell)\in A'} f'_{1,\ell} = P - d^-(1), \tag{26}$$

$$f'_{i,j} \geqslant 0 \quad \text{integer} \quad \forall (i,j) \in A'. \tag{27}$$

Example. Consider a graph such as the one depicted in Fig. 2, with the represented deadhead arcs and associated costs.

Assume that from the demand graph it is known that nodes 2 and 5 are origins with supply equal to 1, node 3 is a transshipment node and 4 is a destination node with demand equal to 3.

From constraints (17) for $i = 1, n+1$, (18) and (19) of (B1), some additional arcs are needed both into and out of nodes 1 and $n+1$. With $P = 2$ vehicles and $d^-(1) = 1 > 0 = d^+(1)$, at least $P - d^+(1) = 2$ additional arcs are needed out of node 1 and at least $P - d^-(1) = 1$ additional arc into 1. So node 1 simultaneously represents an origin with supply 2 and a destination with demand 1. Assuming $K = 5$, node $n+1$ will represent an origin and a destination with supply and demand equal to $K = 5$.

To solve a flow problem (B1′) equivalent to (B1) graph G is transformed into a graph G' where each node represents an origin, or a transshipment node, or a destination (see Fig. 3).

Problem (B1′) can now be transformed into an equivalent transportation problem if we consider:

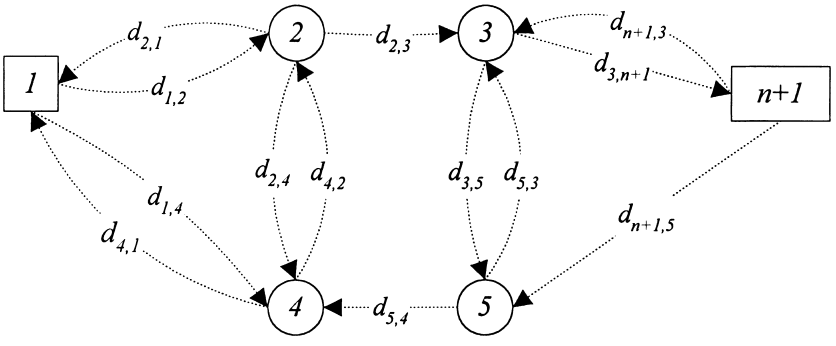


Fig. 2. Deadheading arcs of graph G.

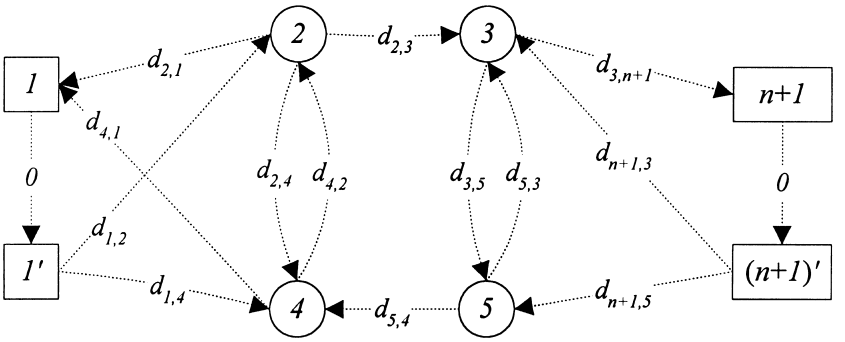


Fig. 3. Deadheading arcs of graph G'.

- as origins (node set O), nodes:
  $i \in U: D(i) > 0$, with supply $a(i) = D(i)$, since they are origins of (B1′) by Eq. (22);
  $(n+1)'$, with supply $a\big((n+1)'\big) = K$, by Eq. (24);
  $1'$, with supply $a(1') = P - d^+(1)$, by Eq. (25).
- as destinations (node set D), nodes:
  $j \in U: D(j) < 0$, with demand $b(j) = -D(j)$, by conditions (22);
  $n+1$, with demand $b(n+1) = K$, by Eq. (23);
  1, with demand $b(1) = P - d^-(1)$, by Eq. (26).
- the unit transportation costs, from each origin to each destination, defined by

$$l_{i,j} = \begin{cases} scl(1) & \text{if } i = 1' \wedge j = 1, \\ scl(n+1) & \text{if } i = (n+1)' \wedge j = n+1, \\ spl(i,j) & \text{otherwise.} \end{cases}$$

These costs are the minimum cost path lengths from each source to each sink of G', since it can be shown that the minimum cost path length from $1'$ to 1 (and from node $(n+1)'$ to $(n+1)$), in G', is equal to the minimum cost circuit length containing node *1* $(n+1)$, in G. It is therefore possible to solve an equivalent transportation problem instead of solving (B1′).

It is now easily shown that a valid lower bound to the optimal value of our problem can be computed as:

$$\text{LB1} = C_{A_R} + z^*_{\text{TP1}},$$

where z^*_{TP1} is the optimal value of the following transportation problem:

$$(\text{TP1}) \quad \min\ z_{\text{TP1}} = \sum_{j \in D} \sum_{i \in O} l_{i,j} t_{i,j},$$

s.t. :

$$\sum_{j \in D} t_{i,j} = a(i) \quad \forall i \in O,$$

$$\sum_{i \in O} t_{i,j} = b(j) \quad \forall j \in D,$$

$$t_{i,j} \geqslant 0 \quad \forall i \in O,\ j \in D.$$

Each variable, $t_{i,j}$, represents the number of times that a minimum cost path from i to j (or the minimum cost circuit, if i and j represent the same initial node) must be deadheaded, in the original problem.

3.2. Method LB2

When the demand graph, G_R, is not connected the solution multigraph obtained with method LB1 may have several connected components. With method LB2 we attempt to reduce the number of connected components in the solution introducing new connectivity constraints.

Suppose that $G_R = (V, A_R)$ is not connected and that at least one of its components, say $G_R^\ell = \left(V^\ell, A_R^\ell\right)$ is balanced, i.e. $d^-(i) = d^+(i)\ \forall i \in V^\ell$.

For such a component we know from constraints (13) that

$$\sum_{r=1}^{L} \sum_{(i,j)\in A_R^\ell} q_{i,j}\, x_{i,j}^r \leqslant W \sum_{r=1}^{L} \sum_{\substack{(i,j)\in A \\ i\in V^\ell, j\notin V^\ell}} y_{i,j}^r.$$

Representing the total demand in G_R^ℓ by Q_ℓ and setting $K_\ell = \lceil Q_\ell / W \rceil$ we get the new constraint

$$\sum_{\substack{(i,j)\in A \\ i\in V^\ell, j\notin V^\ell}} f_{i,j} \geqslant\ K_\ell.$$

If G_R contains nc balanced components, G_R^ℓ $(\ell = 1, \ldots, nc)$, the rationale may be applied to each of them to generate nc connectivity constraints.

To build a minimum cost flow problem including these new connectivity constraints we must be able to transform them into node balance type constraints as we did for constraints (18) and (19).

Let us shrink each balanced component G_R^ℓ into a supernode s^ℓ $(\ell = 1, \ldots, nc)$. Then let us replace

- each arc $(i,j) \in A$ with $i \in V^\ell$ and $j \notin V^\ell$ by an arc (s^ℓ, j) with cost $d_{s^\ell,j} = \min\left\{d_{k,j}\colon\ k \in V^\ell,\ (k,j) \in A\right\}$;
- and each arc $(i,j) \in A$ with $i \notin V^\ell$ and $j \in V^\ell$ by an arc (i, s^ℓ) with cost $d_{i,s^\ell} = \min\{d_{i,k}\colon\ k \in V^\ell,\ (i,k) \in A\}$.

After doing that for all supernodes we have a new graph $\hat{G} = \left(\hat{V},\ \hat{A}\right)$ with $\hat{V} = \left(V \setminus \bigcup_{\ell=1}^{nc} V^\ell\right) \cup \left\{s^1, \ldots, s^{nc}\right\}$.

Applying to $\hat{G}$ the rationale described in Section 3.1 we successively build a standard minimum cost flow problem (B2′) and an equivalent transportation problem (TP2). In (TP2) each supernode s^ℓ $(\ell = 1, \ldots, nc)$ is represented by an origin with supply equal to K_ℓ and by a destination with demand equal to K_ℓ.

With the cost definition given above it follows immediately that

$$\mathrm{LB2} = C_{A_R} + z^*_{\mathrm{TP2}}$$

is a new lower bound on the optimal value of the refuse collection problem. As all nodes in each set V^ℓ $(\ell = 1, \ldots, nc)$ are represented in $\hat{G}$ by a single supernode, s^ℓ, in subproblem (B2′) we are not able to impose node balance constraints on individual nodes in V^ℓ $(\ell = 1, \ldots, nc)$. As a consequence LB2 does not dominate LB1.

When just one balanced component exists in G_R, i.e. $nc = 1$, it can be shown in Ref. [9] that

$$z^*_{\mathrm{PT2}} - K_1\, scl\left(s^1\right) \leqslant z^*_{\mathrm{PT1}} \leqslant z^*_{\mathrm{PT2}} + K_1\, L_{\max}\left(s^1\right),$$

where $scl\,(s^1)$ is taken in $\hat{G}$ and $L_{\max}\,(s^1)$ is the cost of the highest minimum cost path between a pair of nodes in V^1.

4. Heuristic method

4.1. Method HM

The heuristic we developed is a route-first, cluster-second method. The procedure is based on the transformation of the solution (usually unfeasible) obtained with the lower-bounding method LB1, into a near-optimal solution.

Observe that the multigraph solution of the lower-bounding procedure,

$$G_{\mathrm{LB1}} = (V, A_R \cup A_{\mathrm{TP1}}),$$

with A_{TP1} representing the arc set associated to an optimal transportation solution, is Eulerian. It is therefore possible to generate one or more giant circuits in G_{LB1} by identifying one or several Eulerian circuits in this multigraph. By breaking down these giant circuits we obtain a set of feasible trips, i.e., trips from 1 to $n+1$, from $n+1$ to 1, or from $n+1$ to $n+1$, where the vehicle's capacity is not exceeded. These trips are then used to generate a set of P tours, one for each vehicle. Generically the heuristic method can be described as follows.

Heuristic: HM
0. Let G_{LB1} be the multigraph solution of the lower-bounding procedure with
   nc connected components, i.e., $G_{\mathrm{LB1}} = \bigcup_{k=1}^{nc} G^k_{\mathrm{LB1}}$;
   Set $k \leftarrow 1$;
Phase One {Generation of a set of feasible trips}
1. Consider G^k_{LB1};
   1.1. {Built a giant tour in G^k_{LB1}}
        Find an Eulerian Circuit (EC) in G^k_{LB1};
   1.2. Construct a set of feasible trips from EC
        Follow EC as long as the vehicle's capacity allows and either node 1
        or node $n+1$ are not met;
        Break down EC and construct a feasible trip (from $n+1$ to $n+1$;
        from $n+1$ to 1 or from 1 to $n+1$);
   1.3. *if* (all demand arcs of G^k_{LB1} have been considered)
        then goto 1.4.
        else goto 1.2.
   1.4. $k \leftarrow k+1$
        if $k \leqslant nc$ *then goto* 1.
Phase Two
2. Patch trips together if it yields a lower cost set of feasible trips;
Phase Three {Generation of a feasible set of P trips}
   3.1. Change, if needed, the starting node of some of the trips in order to get
        a set of trips with P initial trips;
   3.2. Change, if needed, the final node of some of the trips in order to get
        a set of trips with P final trips.

4.2. Performance analysis

The cost of the near-optimal solution generated by the heuristic, UB, is an upper-bound on the optimal cost for the refuse collection problem. An worst-case bound on the relative gap between UB and the

lower bound LB1 value is given in the next theorem. In an attempt to simplify the proof, we have considered that the multigraph associated to the lower-bounding solution is formed by one connected component only. The result can be extended to the general case, when this multigraph is formed by *nc* connected components.

Theorem. *Let LB1 and UB be the lower and upper bound values, computed by the methods presented in this paper. Suppose that* G_{LB1} *has only one connected component.*

(a) *If* $P \geqslant \max\{d^-(1); d^+(1)\}$, *then*

$$\frac{\mathrm{UB}-\mathrm{LB1}}{\mathrm{LB1}} \leqslant \frac{(L+P-2)lcl(n+1)}{C_{A_R}+K\,scl(n+1)}. \tag{28}$$

(b) *If* $P < \max\{d^-(1);\ d^+(1)\}$, *then*

$$\frac{\mathrm{UB}-\mathrm{LB1}}{\mathrm{LB1}} \leqslant \frac{\left(L+\overline{d}-2\right)\ lcl(n+1)\ +\left(\overline{d}-P\right)\left(spl(1,n+1)+spl(n+1,1)\right)}{C_{A_R}+K\,scl(n+1)}$$

with $\overline{d} = \max\{d^-(1);\ d^+(1)\}$.

Proof. Let EC be an Eulerian Circuit in the multigraph associated with the solution given by method LB1.

(a) To generate a feasible solution, in phase one, we follow EC, from node $n+1$, as long as nodes $n+1$ and 1 are not hit and the residual vehicle's capacity is not smaller then the demand of the following demand arc. Nodes $n+1$ and 1 are hit K and P times, respectively. When node $n+1$ is hit the sequence of arcs in EC is kept (we just finish the current trip and initiate a new one). No cost increase is therefore incurred. The same applies to the first time node 1 is hit. In the worst case, with node 1 being hit the remainder $P-1$ times, (i) EC must be abandoned, (ii) a circuit with cost $lcl(n+1)$ included into the solution, and (iii) EC then resumed, with a total cost increase of $(P-1)\ lcl(n+1)$.

When the vehicle capacity would be exceeded by including the next demand arc in EC a circuit with a worst case cost of $lcl(n+1)$ must again be included in the solution. This happen at most $L-1$ times. So at end of phase one the total cost is, in the worst case, $\mathrm{LB1}+(P+L-2)\ lcl(n+1)$.

In the worst case, no patching of trips is performed in the second phase of the heuristic, thus remaining unchanged the value of the generated solution.

As $P \geqslant \max\{d^-(1);\ d^+(1)\}$ there exist no more than P initial trips and P final trips when entering phase three. This leads to conclude that, in this phase, there will never exist changes of initial and final nodes of any trips which might lead to an increase of the solution value.

So,

$$\mathrm{UB} \leqslant \mathrm{LB1}+(L+P-2)lcl(n+1) \Rightarrow \frac{\mathrm{UB}-\mathrm{LB1}}{\mathrm{LB1}} \leqslant \frac{(L+P-2)lcl(n+1)}{\mathrm{LB1}}$$

Since,

$$\mathrm{LB1} \geqslant C_{A_R}+Kscl(n+1),$$

we get

$$\frac{\mathrm{UB}-\mathrm{LB1}}{\mathrm{LB1}} \leqslant \frac{(L+P-2)lcl(n+1)}{C_{A_R}+K\,scl(n+1)}.$$

(b) If $P < \max\{d^-(1);\ d^+(1)\}$ then in phase one node 1 is hit $\bar{d}$ times $\left(\overline{d} = \max\{d^-(1);\ d^+(1)\}\right)$ rather then P times yielding, in the worst case, a cost increase of $\left(\overline{d}-1\right)\ lcl(n+1)$. Again, in the worst case, in

phase two no patching of trips yields a cost decrease. In phase three $(\overline{d} - P)$ changes will be required in steps 3.1. and 3.2. with a worst case cost increase of

$$(\overline{d} - P)\ (spl(1, n+1) + spl(n+1, 1)).$$

So the worst case gap is

$$\frac{\mathrm{UB} - \mathrm{LB1}}{\mathrm{LB1}} \leqslant \frac{(L + \overline{d} - 2)\ lcl(n+1)\ + (\overline{d} - P)\,(spl(1, n+1) + spl(n+1, 1))}{C_{A_R} + K\ scl(n+1)}. \qquad \square$$

4.3. Computational experience

We also made an empirical comparison of the results given by the heuristic and by method LB1 on a set of test problems. We generated the test problems according to our knowledge about the real world instances. Details of the generation procedures are given in the appendix.

Table 1
Computational results

$n+1$	W	Q_T	$\lvert A_R\rvert$	$K+P$	No. of trips	Worst gap (%)	Gap (%)	% of trips occupying			
								0–30% W	30–50% W	50–70% W	70–100% W
10	140	935	16	7.0	8.0	90	15.7	0	5	0	95
	150	851	14	6.2	7.2	94	18.2	0	3	5	92
	180	910	15	5.0	6.4	140	23.1	0	9	22	69
	200	793	13	4.0	4.6	127	17.0	4	4	4	88
	210	830	14	4.2	5.0	139	19.1	0	4	8	88
17	180	1601	27	9.0	10.8	106	17.6	0	4	17	79
	200	1896	31	10.0	11.6	140	14.7	0	2	14	84
	230	1799	30	8.0	9.6	128	20.4	4	0	4	92
	260	1780	29	7.0	7.8	153	13.2	0	5	3	92
	340	1295	27	5.2	6.0	154	14.5	3	10	3	84
26	280	2706	44	10.0	11.4	139	15.6	2	5	2	91
	300	2861	47	10.6	11.4	168	9.2	4	0	0	96
	320	2804	46	9.0	10.0	125	13.3	0	2	8	90
	340	2871	47	9.0	10.0	137	14.1	8	2	2	88
	370	2904	47	8.0	9.0	150	14.0	2	0	9	89
	460	2709	44	6.0	7.0	140	16.5	9	6	0	85
37	420	4196	68	10.0	11.0	83	10.9	0	2	2	96
	460	4384	71	9.6	10.6	171	11.6	0	4	7	89
	510	4591	74	9.0	10.0	154	14.2	0	0	10	90
	530	4176	68	8.0	9.0	149	12.7	0	9	0	91
	620	4298	70	7.0	8.2	122	14.8	0	7	15	78
	670	3964	64	6.0	7.0	154	15.1	0	0	3	97
	720	4282	69	6.2	7.0	115	12.5	3	11	3	83
50	580	5802	95	10.0	11.0	116	11.5	0	7	2	91
	650	5808	95	9.2	10.0	153	9.7	0	6	6	88
	730	5797	95	8.2	9.0	142	11.3	0	7	7	86
	750	5659	93	7.6	8.6	122	12.3	0	7	7	86
	810	5665	93	7.2	8.0	158	11.0	5	2	4	83
	860	5982	97	7.2	8.0	147	10.6	2	8	5	85
	960	5538	94	6.0	7.0	126	14.5	8	6	0	86

Computational results were obtained with an Intel Pentium processor @100 MHz, configured with 24 MB of memory, running Windows 95. The programming language used was Pascal using the Turbo–Pascal compiler.

The processing CPU times spent for the test calculations are insignificant and as such were not taken into account.

We define the gap, as usual, by gap $= \frac{\text{UB}-\text{LB}}{\text{LB}} \times 100$.

Every row in Table 1 shows the average values of ten test problems with similar characteristics. For every dimension, vehicles with different capacities were considered, along with a different number of demand arcs. The number of demand arcs, $|A_R|$, represents approximately 80% of the total number of arcs on the graph G, $|A|$.

The values in column 7, "Worst gap", represent the right-hand side of expression (28) depicted as a percentage. As expected the gap values obtained are far away from their worst case value and seem to decrease with the number of nodes. The number of trips generated is usually close to the theoretical lower bound on the number of trips in any feasible solution we deduced in Section 3. $(K + P)$, indicating that, on average, vehicle capacity is efficiently used. This idea is confirmed by figures in the last 4 columns of the table.

5. Final remarks

The lower bounding values and the heuristic solution for the refuse collection problem presented in this paper are all based on a transportation model. They are very fast to compute and produced a relative gap of moderate size in our computational study. To reduce the gap value we are now trying to develop a new lower bounding method based on a minimum cost flow problem coupled with Lagrangean relaxation and testing new heuristic algorithms based on the cluster first/route second approach.

Appendix A

We first generate a sparse, planar, connected and undirected graph. We then transform this graph into a directed one and select a subset of the arcs to represent the demand arcs. To generate the problem's input data we begin by randomly obtaining the deadheading costs. Arc demands and service costs are a function of the deadheading costs, as it is next described. As in the real case we set the number of vehicles to 2.

- Deadheading costs

  First consider $d_{i,j} = \infty,\ \forall (i, j) \in A$.

  Given L1, $c_1 \in (0, 1)$ and $c_2 \in (0, 1)$ the deadheading cost of an arc (i, j), not incident to $n + 1$,

$$d_{i,j} \text{ is a } \begin{cases} \text{uniform random number over } [L1 \mp c_1 L1] & \text{if } d_{j,i} = \infty, \\ \text{uniform random number over } \left[d_{j,i} \mp c_2 d_{j,i}\right] & \text{if } d_{j,i} < \infty. \end{cases}$$

  The deadheading cost of an arc (i, j), incident to $n + 1$,

$$d_{i,j} \text{ is a } \begin{cases} \text{uniform random number over } [(n+1)L1 \mp c_1 L1] & \text{if } d_{j,i} = \infty, \\ \text{uniform random number over } \left[d_{j,i} \mp c_2 d_{j,i}\right] & \text{if } d_{j,i} < \infty. \end{cases}$$

  In our test problems we let L1 assume integer values between 15 and 20. We assume two values for $c_1 = 0.1;\ 0.25$ and set $c_2 = 0.05$.

- *Arc demands*

  Given $c_4,\ c_5 > 0$, $q_{i,j}$, the demand of each arc $(i, j) \in A_R$ is an uniform random number over $\left[c_4\, d_{i,j} \mp c_5\right]$. We set $c_4 = 4$ and $c_5 = 1$.

- *Service costs*

  Given c_6, $c_7 > 0$, $c_{i,j}$, the service cost of each arc $(i, j) \in A_R$ is an uniform random number over $\left[c_6\, q_{i,j} \mp c_7\right]$. We set $c_6 = 0.5$ and $c_7 = 1$.

References

[1] A.A. Assad, B.L. Golden, Arc routing methods and Applications. in: M.O. Ball, T.L. Magnanti, C.L. Monma, G.L. Nemhauser (Eds.), Handbooks in Operations Research and Management Science, Vol. 8, Network Routing, Elsevier, Amsterdam, 1995, pp. 375–483.
[2] E.J. Beltrami, L.D. Bodin, Networks and vehicle routing for municipal waste collection, Networks 4 (1974) 65–94.
[3] L.D. Bodin, A taxonomic structure for vehicle routing and scheduling problems, Computers and Urban Society 1 (1975) 11–29.
[4] L.D. Bodin, S.J. Kursh, A detailed description of a computer system for the routing and scheduling of street sweepers, Computers and Operations Research 6 (1979) 181–198.
[5] H.A. Eiselt, M. Gendreau, G. Laporte, Arc routing problems. Part I: The Chinese postman problem, Operations Research 43 (1995) 231–242.
[6] H.A. Eiselt, M. Gendreau, G. Laporte, Arc routing problems. Part II: The rural postman problem, Operations Research 43 (1995) 399–414.
[7] G.N. Frederickson, M.S. Hetch, C.E. Kim, Approximation algorithms for some routing problems, SIAM Journal on Computing 7 (1978) 178–193.
[8] B.L. Golden, R.T. Wong, Capacitated arc routing problems, Networks 11 (1981) 305–315.
[9] M.C. Mourão, Tours optimization in urban waste collection: Models and algorithms, Ph.D. Thesis, ISEG, Technical University of Lisbon, Portugal, 1998 (in Portuguese).